Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*The application of Large Language Models (LLMs) for the interpretation of police and/or insurance reports of traffic accidents and the derivation of the underlying traffic scenario for storage in a scenario database*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (**x**), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (**x**), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		*Mechanical & Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	*Testing / Inspection / Analysis Techniques*	*Automobile / Rail Vehicles*	*Safety Improvement Technologies*
Concept of Proposed Technology	*For the safety assessment of Automated Driving Vehicles, it is important to have a complete view on the scenarios that such vehicle may encounter on the road during its lifetime. An important source of information to determine the relevant scenarios is provided by reports that are made on traffic accident sites. Depending on the severity of the accident, police and/or insurance companies will draft such accident reports. As these reports are not yet standardized, and are written by a variety of police or insurance officers, it is far from straightforward to derive the scenario that has led to the accident from this report.* *It is proposed to train Large Language Models (LLMs) for interpretation of accidents reports and derivation of the scenarios that resulted in the accidents. The resulting scenarios are added to a scenario database that consists of representative scenarios for a certain traffic system. Scenarios are used to describe the Operational Design Domain of AD vehicles and to describe test scenarios for the safety assessment of AD vehicles.*			

Rationale for Joint R&D Project	*Methods to describe traffic as a sequence of scenarios have been developed over the last decade, and these methods are readily available. Following the New Assessment/Test Method as proposed by UNECE (1), such scenarios are used for the description of the Operational Design Domain of AD vehicles, and for the generation of test scenarios for safety assessment. It is essential for a good assessment of vehicles by the respective authorities (e.g. KATRI, KOROAD, and the Police Department to approve the deployment of such vehicle), that the scenarios collected in the scenario database provide a representative view of the scenarios that the vehicle under test may encounter during its lifetime.* *Analysis methods to identify and characterize scenarios from state-of-the-art sensor sets on board of vehicles are available (2). Though it is important to collect such scenarios from vehicles in naturalistic driving studies, these studies usually do not provide information on typical accidents that occur. In the absence of an in-depth accident database, access to this type of information can be derived from police or insurance companies that report traffic accidents. For national accident statistics not a high level of detail is needed from these reports. However, to derive the scenario that took place that resulted in the accident, human interpretation of such (police) reports is required, and this is very time consuming as each report needs to be studied separately.* *With this project, we would like to develop automatic interpretation tools for this type of reports. The tools should be able to derive the most likely scenario from the accident report and store the scenario with appropriate meta data into a scenario database. LLMs are believed to be specifically adequate for interpreting such reports.* *Type approval authorities in countries that have a focus on deploying AD vehicles on their roads (such as South Korea and the Netherlands), do have a need for detailed accident analyses, and for collecting scenarios not only from naturalistic driving, but from traffic accidents as well. It is for this reason that international collaboration is required. The outcomes of the research proposed in this project are expected to provide an accident scenario database and guidelines for authorities and organizations on how to report accidents to increase the reliability and applicability of information resulting from such reports.*
Key Words	1. Scenario-based safety assessment 2. Automated Driving 3. In-depth accident reports 4. Application of Large Language Models (LLMs) 5. Scenario identification and parametrization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (1)	End of the Project	Level (6)
R&D Objectives	• *The application of Large Language Models for the interpretation of (police) traffic accident reports.* • *Scenario mining techniques from information as collected in such reports and storage of the resulting scenarios with the appropriate meta information in a scenario database.* • *A methodology to convert the stored accident scenarios into test scenarios for the testing of AD vehicles.*			
R&D Target Specifications	• *The target is to automatically analyze 100 (police) accident reports from 3 different regions and store the resulting scenarios in a specific Korean scenario database.* • *In a similar way, 100 police accident reports originating from the Netherlands will be analyzed using the same algorithms (possibly with other parameter settings).* • *Show easy interfacing with reports as input and scenarios in a scenario database as output.* • *The demonstration should reveal what additional development needs to be done to scale up the automatic analysis of accident reports that are collected from all different regions in South Korea.* *Availability of such reports is a risk. In the Netherlands, the reports are collected under the responsibility of the Ministry of Justice and Security. It might be possible to get access to such reports for R&D purposes and when all data are anonymized. It needs to be investigated how the availability is of such reports in South Korea.*			
R&D Plan	***WP1 Analyze types of accident police reports*** • *Investigate different types of accident reports from various police sources to check the level of detail and type of information available. The investigation is performed for both **South Korea** and for **the Netherlands**.* • *Overview of information that is available to be reported from accident reports in Korea and the Netherlands.* • *Select a variety of typically different police accident reports as basis for the current study.* ***WP2 Evaluate LLMs to derive pre-crash scenarios from police report data*** • *Study the use of LLMs to automatically interpret police reports, incl. sketches made of the accident location.* • *Develop a supervised methodology for police accident report interpretation that is supported by LLMs.* • *Evaluate the developed methodology by interpreting the variety of police reports selected in WP1.* ***WP3 Develop LLMs to derive pre-crash scenarios from police report data*** • *Further develop LLM technology for the interpretation of police accident reports.* • *Study the added value of **virtual simulation** to enhance the reliability of data* • *Provide a standardized data format to compile the enhanced data and (meta-)information of pre-crash scenarios and store these into a pre-crash scenario database.* ***WP4 Set up a pre-crash scenario database*** • *Provide a framework and governance system to store pre-crash scenarios derived from police accident reports into a national pre-crash scenario database.* • *Select data formats that ensure linking the pre-crash scenario database to the more general national scenario database for the assessment of automated driving systems.* • *Demonstrate the developed methodology by interpreting 100 Korean and 100 Dutch*			

	police accident reports and storing the results in respectively a Korean and Dutch pre-crash scenario database. ***WP5 Dissemination and exploitation*** • *Connect to the National Authorities to establish a formal process for collecting, interpreting and storing pre-crash scenarios from police accident reports in national databases.* • *Provide training to relevant authority officials and police officers to standardize and enhance the information collected in police accident reports, to support road safety analysis on a national level.* • *Publish the results of the project to support international standardization.*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Coordinate the project* • *Establish connections to relevant stakeholders in the Netherlands / the EU such as, e.g., SWOV, Min. of IenW, Min. of JenV, TASC initiative, etc.* • *Establish connections to relevant stakeholders in Korea, such as, e.g., KoRoad, MORAI, KATECH, etc.* • *Collect and analyze police reports from the Netherlands* • *Develop and apply LLM technology for the analysis of Dutch police reports* • *Provide formats for a scenario database and establish the Dutch pre-crash scenario database* • *Provide a formal process for establishing a national pre-crash scenario database* • *Set up a training schedule for authority officials and police officers on collecting information for use in scenario databases* • *Provide training and education of relevant Dutch authority officials and police officers* • *Write one or more publications to disseminate the results of the project*
Expected R&R of Korean SME Partner	• *Develop methods for the interpretation of police accident reports (text, notes, sketches, photographs, etc.)* • *Develop LLMs to support the interpretation of police accident reports* • *Develop mechanism for data transfer and data storage to collect this type of information at a central location* • *Develop and set up a scenario database for Korea, following formats and experiences provided by TNO* • *Develop supporting technologies, such as virtual simulation (MORAI), to enhance data collected from police accident reports* • *Support the development of training material and the training and education of relevant Korean authority officials and police officers*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *A comprehensive accident scenario database used both for general road safety statistical analyses and to aid in scenario-based evaluation of automated driving systems and vehicle safety features.* • *A system for ongoing collection, interpretation, and storage of police accident report data in machine-readable format in both Korea and the Netherlands. This mechanism may support international standardization efforts.* • *The increased use of LLMs for interpreting a diversity of human-written reports – including text, notes, images, and graphs – should also help standardize accident reporting formats for police officers.*

6. References

(1) UNECE WP29 GRVA, "New Assessment/Test Method for Automated Driving (NATM)-Master Document," UNECE, Geneva, Switzerland, 2021.

(2) de Gelder, Erwin; Op den Camp, Olaf; Broos, Jeroen; Paardekooper, Jan-Pieter; Uittenbogaard, Jeroen; van Montfort, Sjef; Kalisvaart, Sytze; Goossens, Henk "TNO StreetWise - Scenario-based safety validation of Automated Driving Systems", TNO, www.tno.nl/StreetWise, 2023.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization